maintask

February 18, 2026

[1]:
```
!pip install skorch
```

```
Collecting skorch
  Downloading skorch-1.3.1-py3-none-any.whl.metadata (11 kB)
Requirement already satisfied: numpy>=1.13.3 in /usr/local/lib/python3.12/dist-
packages (from skorch) (2.0.2)
Requirement already satisfied: scikit-learn>=0.22.0 in
/usr/local/lib/python3.12/dist-packages (from skorch) (1.6.1)
Requirement already satisfied: scipy>=1.1.0 in /usr/local/lib/python3.12/dist-
packages (from skorch) (1.16.3)
Requirement already satisfied: tabulate>=0.7.7 in
/usr/local/lib/python3.12/dist-packages (from skorch) (0.9.0)
Requirement already satisfied: tqdm>=4.14.0 in /usr/local/lib/python3.12/dist-
packages (from skorch) (4.67.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-
packages (from scikit-learn>=0.22.0->skorch) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in
/usr/local/lib/python3.12/dist-packages (from scikit-learn>=0.22.0->skorch)
(3.6.0)
Downloading skorch-1.3.1-py3-none-any.whl (268 kB)
                         0.0/268.5 kB
? eta -:--:--
                      268.5/268.5 kB
13.9 MB/s eta 0:00:00
Installing collected packages: skorch
Successfully installed skorch-1.3.1
```

[2]:
```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

[3]:
```
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd

import pickle
```

```
from sklearn.metrics import mean_squared_error
from sklearn.model_selection import StratifiedKFold, GridSearchCV

from skorch import NeuralNetRegressor
from skorch.callbacks import ProgressBar, LRScheduler, Checkpoint,␣
↪EarlyStopping, EpochScoring
from skorch.helper import SliceDict

import torch
import torch.nn as nn

from transformers import BertTokenizer, BertModel
```

[4]:
```
# ===== CONFIG =====
SEED = 42
MAX_LEN = 128
DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
WEIGHTS_PATH = "/content/drive/MyDrive/GSIdetect/bert_resize/weights.pth" #"/
↪content/drive/MyDrive/Colab Notebooks/weights.pth"
DATA_PATH = "/content/drive/MyDrive/GSIdetect/dati/train.pkl" #"/content/drive/
↪MyDrive/Colab Notebooks/train.pkl"  # _dataset_completo.pkl"
TOKENIZER_PATH = "/content/drive/MyDrive/GSIdetect/bert_resize" #"/content/
↪drive/MyDrive/Colab Notebooks/tokenizer"

torch.manual_seed(SEED)
np.random.seed(SEED)
```

1 Obiettivo dell'esperimento

L'obiettivo di questo notebook è affrontare un problema di **regressione su testo in lingua italiana** nell'ambito della challenge EVALITA GSI:detect (Detecting Gender Stereotypes in Italian) utilizzando un modello BERT pre-addestrato, confrontando due strategie:

- **Feature extraction** (BERT congelato)
- **Fine-tuning completo** del modello

Il confronto è effettuato in termini di **Mean Squared Error (MSE)**, utilizzando cross-validation stratificata per garantire stabilità statistica rispetto alla distribuzione delle categorie.

[5]:
```
tokenizer = BertTokenizer.from_pretrained(TOKENIZER_PATH)
print(f"len(tokenizer)={len(tokenizer)}")
```

```
len(tokenizer)=31197
```

[6]:
```
with open(DATA_PATH, "rb") as f:
    data = pickle.load(f)
```

```
texts = data["text"]
values = data["gs_value"]
categories = data["gs_category"]
```

[7]:
```
def prepare_data(text_series, tokenizer, max_len):
    enc = tokenizer(
        text_series.tolist(),
        truncation=True,
        padding="max_length",
        max_length=max_len,
        return_tensors="np"
    )
    X = SliceDict(
        input_ids=enc["input_ids"],
        attention_mask=enc["attention_mask"]
    )
    return X
```

[8]:
```
X = prepare_data(texts, tokenizer, MAX_LEN)
y = values.values.astype(np.float32).reshape(-1, 1)
print(f"len(X)={len(X)}, len(y)={len(y)}")
```

```
len(X)=382, len(y)=382
```

1.1 Baseline di riferimento

Prima di addestrare modelli complessi, viene calcolata una baseline utilizzando un DummyRegressor che predice sempre la media.

Un modello che non supera tale valore è da considerarsi non informativo, ma un modello che lo supera è ancora da intendersi come potenziale candidato e non necessariamente ottimale per gli obiettivi della challenge.

[9]:
```
from sklearn.dummy import DummyRegressor
from sklearn.metrics import mean_squared_error

dummy = DummyRegressor(strategy="mean")
dummy.fit(X, y)
preds = dummy.predict(X)
baseline_mse = mean_squared_error(y, preds)

print(f"MSE se predicessi sempre la media: {baseline_mse:.4f}")
```

```
MSE se predicessi sempre la media: 0.1412
```

1.2 Strategia di validazione

La valutazione è effettuata tramite cross-validation stratificata rispetto alle categorie del dataset (`gs_category`), al fine di preservare la distribuzione dei dati nei vari split e ridurre la varianza della stima dell'errore.

```
[10]: y_cats = categories.values
skf = StratifiedKFold(n_splits=3, shuffle=True, random_state=42)
cv_splits = list(skf.split(X, y_cats))
```

2 Scelta del modello linguistico

Si utilizza `dbmdz/bert-base-italian-xxl-cased`, un modello BERT addestrato specificamente su corpora italiani.

Il modello viene adattato a regressione tramite:

- una **testa fully-connected dedicata**
- **mean pooling sugli embeddings contestuali**

2.1 Testa di regressione

Gli embeddings pooled vengono passati a una rete fully-connected con riduzione progressiva della dimensionalità.

La testa fully-connected è progettata per:

- ridurre progressivamente la dimensionalità
- migliorare la stabilità con BatchNorm
- limitare l'overfitting tramite Dropout, soprattutto in un caso come questo con pochi dati di addestramento

L'output è vincolato in [0, 1] tramite funzione Sigmoid, in accordo con il dominio del target della challenge.

2.2 Pooling degli embeddings

Invece di utilizzare il solo token `[CLS]`, viene adottata una strategia di **mean pooling pesato sull'attention mask**.

Questa scelta consente di:

- incorporare l'informazione di tutti i token rilevanti
- ottenere rappresentazioni più stabili per task di regressione
- ridurre la dipendenza da un singolo token speciale

```
[11]: class BertRegressor(nn.Module):
    def __init__(self, freeze_bert, tokenizer_len):
        super().__init__()
        self.bert = BertModel.from_pretrained("dbmdz/
↪bert-base-italian-xxl-cased")
        if tokenizer_len:
            self.bert.resize_token_embeddings(tokenizer_len)
        try:
            state_dict = torch.load(WEIGHTS_PATH, map_location="cpu")
```

```
            state_dict_fixed = {k.replace('bert.', '', 1): v for k, v in␣
↪state_dict.items()}
            self.bert.load_state_dict(state_dict_fixed, strict=False)
            print("Pesi caricati.")
        except:
            print("Errore caricamento pesi.")
        if freeze_bert:
            for p in self.bert.parameters():
                p.requires_grad = False
        input_dim = self.bert.config.hidden_size  # 768
        self.regressor = nn.Sequential(
            nn.Linear(input_dim, 128),
            nn.BatchNorm1d(128),
            nn.ReLU(),
            nn.Dropout(0.5),
            nn.Linear(128, 1),
            nn.Sigmoid()
        )

    def forward(self, input_ids, attention_mask):
        out = self.bert(input_ids=input_ids, attention_mask=attention_mask)
        token_embeddings = out.last_hidden_state
        input_mask_expanded = attention_mask.unsqueeze(-1).
↪expand(token_embeddings.size()).float()
        sum_embeddings = torch.sum(token_embeddings * input_mask_expanded, 1)
        sum_mask = torch.clamp(input_mask_expanded.sum(1), min=1e-9)
        mean_pooling = sum_embeddings / sum_mask
        return self.regressor(mean_pooling)
```

[12]:
```
risultati = {}
```

3 Perché Skorch

Skorch è utilizzato come interfaccia tra PyTorch e scikit-learn per tre motivi chiave:

1. **Integrazione nativa e diretta con `GridSearchCV`, come suggerito nella documentazione**
2. **Gestione automatica di training / validation / callbacks**
3. **Riproducibilità e tracciamento delle metriche per epoca**

Questo approccio permette di trattare modelli deep learning come stimatori sklearn, rendendo il confronto sperimentale più rigoroso e standardizzato (`fit`, `best_params_`, `best_score_`, `best_estimator_`, `best_estimator_.history`, …).

Il notebook del task di classificazione segue invece il classico approccio con train_loop e test_loop in puro torch.

Osservazione: poiché la GridSearch è progettata per massimizzare i punteggi, utilizziamo `neg_mean_squared_error` per trasformare l'errore in un valore negativo (dove il numero 'più alto' corrisponde all'errore minore) e applichiamo il segno meno finale per visualizzare il reale MSE positivo.

[13]:
```
def addestra(freeze_bert, X, y):
    print(f"freeze_bert={freeze_bert}")

    cp = Checkpoint(monitor='valid_loss_best', dirname=f'/content/drive/MyDrive/
 ↪Colab Notebooks/exp_freeze_{freeze_bert}', load_best=True)
    es = EarlyStopping(monitor='valid_loss', patience=5)
    scheduler = LRScheduler(policy='ReduceLROnPlateau', monitor='valid_loss',
                            patience=3, factor=0.5)

    net = NeuralNetRegressor(
        module=BertRegressor,
        module__tokenizer_len=len(tokenizer),
        module__freeze_bert=freeze_bert,
        criterion=nn.MSELoss,
        optimizer=torch.optim.AdamW,
        optimizer__weight_decay=0.01,
        device=DEVICE,
        max_epochs=30,
        iterator_train__shuffle=True,
        iterator_train__drop_last=True,
        callbacks=[cp, es, scheduler],
        verbose=1,
    )

    params = {
        'optimizer__lr': [1e-3, 5e-4] if freeze_bert else [1e-5, 2e-5],
        'batch_size':      [32, 64]   if freeze_bert else [16]
    }

    gs = GridSearchCV(
        net,
        params,
        refit=True,
        cv=cv_splits,
        scoring='neg_mean_squared_error',
        verbose=3
    )

    gs.fit(X, y)

    print("Migliori iperparametri:", gs.best_params_)
    print("Miglior MSE:", -gs.best_score_)
```

```
    return gs.best_estimator_, gs.best_estimator_.history
```

4 Confronto tra strategie di addestramento

Vengono confrontate due modalità:

- **Freeze=True**: BERT utilizzato come estrattore di feature
- **Freeze=False**: fine-tuning completo del modello

4.1 Mode 1 - Freeze (Feature Extraction)

- I pesi di BERT sono congelati
- Learning rate più alto per addestrare solo la testa di regressione
- Batch size maggiore, con minore utilizzo di memoria

4.2 Mode 2 - Fine-Tuning

- Tutti i pesi del modello vengono aggiornati
- Learning rate molto basso per evitare catastrophic forgetting
- Batch size ridotto per limiti di memoria GPU

[14]:
```
for freeze_bert in [True, False]:
    best_est, history = addestra(freeze_bert, X, y)
    risultati[freeze_bert] = {
        "model": best_est,
        "history": history,
        "train_loss": history[:, 'train_loss'],
        "valid_loss": history[:, 'valid_loss']
    }
    best_est.save_params(f_params=f"/content/drive/MyDrive/Colab Notebooks/
↪regression_freeze_{freeze_bert}.pkl")
```

```
freeze_bert=True
Fitting 3 folds for each of 4 candidates, totalling 12 fits

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94:
UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab
(https://huggingface.co/settings/tokens), set it as secret in your Google Colab
and restart your session.
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access
public models or datasets.
  warnings.warn(

config.json:   0%|          | 0.00/433 [00:00<?, ?B/s]

model.safetensors:   0%|          | 0.00/445M [00:00<?, ?B/s]
```

```
Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1435        0.1559     +  0.0010  2.6006
      2        0.0864        0.1493     +  0.0010  1.5002
      3        0.0697        0.1436     +  0.0010  1.4611
      4        0.0622        0.1429     +  0.0010  1.4050
      5        0.0494        0.1442        0.0010  1.6794
      6        0.0436        0.1445        0.0010  1.5871
      7        0.0401        0.1427     +  0.0010  1.6025
      8        0.0401        0.1449        0.0010  1.5284
      9        0.0322        0.1468        0.0010  1.4437
     10        0.0275        0.1503        0.0010  1.4489
     11        0.0269        0.1542        0.0010  1.4573
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 1/3] END batch_size=32, optimizer__lr=0.001;, score=-0.103 total time=
1.4min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
```

```
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1537        0.1644     +  0.0010  1.5284
      2        0.0957        0.1584     +  0.0010  1.4859
      3        0.0738        0.1554     +  0.0010  1.5265
      4        0.0581        0.1552     +  0.0010  1.5564
      5        0.0479        0.1553        0.0010  1.5398
      6        0.0378        0.1591        0.0010  1.4974
      7        0.0393        0.1669        0.0010  1.5071
      8        0.0430        0.1736        0.0010  1.5053
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 2/3] END batch_size=32, optimizer__lr=0.001;, score=-0.110 total time=
27.4s

Warning: You are sending unauthenticated requests to the HF Hub. Please set a
HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated
requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
faster downloads.

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1334        0.1564     +  0.0010  1.5679
      2        0.0852        0.1472     +  0.0010  1.5389
      3        0.0720        0.1395     +  0.0010  1.5931
      4        0.0544        0.1333     +  0.0010  1.5373
      5        0.0499        0.1305     +  0.0010  1.5526
      6        0.0380        0.1309        0.0010  1.6405
      7        0.0386        0.1326        0.0010  1.5861
```

```
      8        0.0333       0.1328       0.0010  1.5724
      9        0.0294       0.1339       0.0010  1.5758
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 3/3] END batch_size=32, optimizer__lr=0.001;, score=-0.113 total time=
32.6s

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |  |
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
cls.seq_relationship.weight                 | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1492        0.1582     +  0.0005  1.6515
      2        0.1228        0.1506     +  0.0005  1.6605
      3        0.0966        0.1447     +  0.0005  1.6694
      4        0.0654        0.1399     +  0.0005  1.6305
      5        0.0658        0.1368     +  0.0005  1.6944
      6        0.0619        0.1386        0.0005  1.6487
      7        0.0559        0.1427        0.0005  1.6810
      8        0.0527        0.1445        0.0005  1.7006
      9        0.0431        0.1431        0.0005  1.7021
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 1/3] END batch_size=32, optimizer__lr=0.0005;, score=-0.114 total time=
36.6s

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |  |
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
```

```
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1425        0.1644     +  0.0005  1.7763
      2        0.1089        0.1610     +  0.0005  1.7721
      3        0.0819        0.1579     +  0.0005  1.7438
      4        0.0693        0.1545     +  0.0005  1.7050
      5        0.0630        0.1544     +  0.0005  1.7450
      6        0.0541        0.1549        0.0005  1.6845
      7        0.0496        0.1574        0.0005  1.6998
      8        0.0447        0.1600        0.0005  1.6936
      9        0.0403        0.1632        0.0005  1.6996
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 2/3] END batch_size=32, optimizer__lr=0.0005;, score=-0.109 total time=
35.8s

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1410        0.1603     +  0.0005  1.7038
      2        0.1136        0.1552     +  0.0005  1.6496
      3        0.0907        0.1496     +  0.0005  1.6837
      4        0.0788        0.1424     +  0.0005  1.6772
      5        0.0667        0.1359     +  0.0005  1.7043
      6        0.0567        0.1325     +  0.0005  1.6916
      7        0.0505        0.1317     +  0.0005  1.6567
      8        0.0471        0.1326        0.0005  1.6796
```

```
      9        0.0386        0.1344        0.0005  1.7139
     10        0.0442        0.1379        0.0005  1.7241
     11        0.0354        0.1389        0.0005  1.7378
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 3/3] END batch_size=32, optimizer__lr=0.0005;, score=-0.113 total time=
42.1s

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |  |
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
cls.seq_relationship.weight                 | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1792        0.1598     +  0.0010  1.8491
      2        0.1263        0.1531     +  0.0010  1.7702
      3        0.0999        0.1485     +  0.0010  1.7682
      4        0.0777        0.1444     +  0.0010  1.7333
      5        0.0593        0.1416     +  0.0010  1.7338
      6        0.0557        0.1391     +  0.0010  1.7431
      7        0.0487        0.1369     +  0.0010  1.8058
      8        0.0413        0.1352     +  0.0010  1.8259
      9        0.0411        0.1321     +  0.0010  1.7551
     10        0.0345        0.1312     +  0.0010  1.7432
     11        0.0312        0.1314        0.0010  1.7625
     12        0.0339        0.1324        0.0010  1.7460
     13        0.0277        0.1345        0.0010  1.7567
     14        0.0279        0.1355        0.0010  1.7677
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 1/3] END batch_size=64, optimizer__lr=0.001;, score=-0.103 total time=
56.3s

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
```

```
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1497        0.1680     +  0.0010  1.7978
      2        0.1053        0.1622     +  0.0010  1.7470
      3        0.0910        0.1586     +  0.0010  1.7757
      4        0.0665        0.1553     +  0.0010  1.7982
      5        0.0542        0.1525     +  0.0010  1.7487
      6        0.0490        0.1504     +  0.0010  1.7568
      7        0.0504        0.1497     +  0.0010  1.8014
      8        0.0379        0.1509        0.0010  1.7919
      9        0.0357        0.1529        0.0010  1.7819
     10        0.0343        0.1557        0.0010  1.7997
     11        0.0274        0.1595        0.0010  1.8141
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 2/3] END batch_size=64, optimizer__lr=0.001;, score=-0.107 total time=
47.7s

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
```

```
      1        0.1470        0.1619     +  0.0010  1.8184
      2        0.0897        0.1559     +  0.0010  1.7627
      3        0.0744        0.1521     +  0.0010  1.7833
      4        0.0638        0.1486     +  0.0010  1.7764
      5        0.0546        0.1443     +  0.0010  1.7250
      6        0.0499        0.1402     +  0.0010  1.7403
      7        0.0464        0.1364     +  0.0010  1.7937
      8        0.0383        0.1341     +  0.0010  1.7395
      9        0.0313        0.1333     +  0.0010  1.8078
     10        0.0320        0.1329     +  0.0010  1.7475
     11        0.0257        0.1329        0.0010  1.8234
     12        0.0260        0.1327     +  0.0010  1.7952
     13        0.0219        0.1339        0.0010  1.7933
     14        0.0237        0.1347        0.0010  1.8258
     15        0.0185        0.1348        0.0010  1.8206
     16        0.0216        0.1353        0.0010  1.8486
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 3/3] END batch_size=64, optimizer__lr=0.001;, score=-0.107 total time=
1.1min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                       | Status     |  |
------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1694        0.1628     +  0.0005  1.8112
      2        0.1289        0.1588     +  0.0005  1.8035
      3        0.1032        0.1544     +  0.0005  1.7481
      4        0.0950        0.1499     +  0.0005  1.7560
      5        0.0748        0.1462     +  0.0005  1.7800
      6        0.0704        0.1425     +  0.0005  1.7284
      7        0.0577        0.1385     +  0.0005  1.7660
      8        0.0556        0.1356     +  0.0005  1.7455
      9        0.0586        0.1333     +  0.0005  1.6987
```

```
     10        0.0469       0.1316     +  0.0005  1.7060
     11        0.0439       0.1314     +  0.0005  1.7227
     12        0.0393       0.1323        0.0005  1.7909
     13        0.0457       0.1336        0.0005  1.7720
     14        0.0410       0.1359        0.0005  1.7912
     15        0.0389       0.1396        0.0005  1.8260
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 1/3] END batch_size=64, optimizer__lr=0.0005;, score=-0.107 total time=
1.0min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1719        0.1710     +  0.0005  1.8630
      2        0.1208        0.1660     +  0.0005  1.8244
      3        0.1077        0.1617     +  0.0005  1.7981
      4        0.0869        0.1588     +  0.0005  1.7861
      5        0.0799        0.1571     +  0.0005  1.7701
      6        0.0692        0.1561     +  0.0005  1.7851
      7        0.0572        0.1545     +  0.0005  1.8014
      8        0.0502        0.1523     +  0.0005  1.7682
      9        0.0509        0.1508     +  0.0005  1.7971
     10        0.0428        0.1499     +  0.0005  1.8466
     11        0.0409        0.1501        0.0005  1.8055
     12        0.0400        0.1514        0.0005  1.8214
     13        0.0370        0.1528        0.0005  1.8207
     14        0.0316        0.1538        0.0005  1.8415
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 2/3] END batch_size=64, optimizer__lr=0.0005;, score=-0.107 total time=
1.1min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]
```

```
BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                       | Status     |  |
------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias  | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight    | UNEXPECTED |  |
cls.predictions.bias                      | UNEXPECTED |  |
cls.seq_relationship.bias                 | UNEXPECTED |  |
cls.predictions.transform.dense.bias      | UNEXPECTED |  |
cls.seq_relationship.weight               | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1542        0.1615     +  0.0005  1.8023
      2        0.1139        0.1584     +  0.0005  1.7344
      3        0.1014        0.1553     +  0.0005  1.7659
      4        0.0868        0.1528     +  0.0005  1.7719
      5        0.0746        0.1507     +  0.0005  1.7299
      6        0.0650        0.1487     +  0.0005  1.7281
      7        0.0575        0.1473     +  0.0005  1.7443
      8        0.0527        0.1464     +  0.0005  1.7504
      9        0.0469        0.1457     +  0.0005  1.8056
     10        0.0508        0.1449     +  0.0005  1.7615
     11        0.0400        0.1448     +  0.0005  1.7722
     12        0.0390        0.1449        0.0005  1.8258
     13        0.0419        0.1455        0.0005  1.7867
     14        0.0335        0.1466        0.0005  1.8276
     15        0.0353        0.1481        0.0005  1.8250
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 3/3] END batch_size=64, optimizer__lr=0.0005;, score=-0.111 total time=
1.0min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                       | Status     |  |
------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias  | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight    | UNEXPECTED |  |
cls.predictions.bias                      | UNEXPECTED |  |
cls.seq_relationship.bias                 | UNEXPECTED |  |
cls.predictions.transform.dense.bias      | UNEXPECTED |  |
cls.seq_relationship.weight               | UNEXPECTED |  |
```

```
Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1467        0.1587     +  0.0010  2.5045
      2        0.1033        0.1491     +  0.0010  2.4476
      3        0.0777        0.1438     +  0.0010  2.4038
      4        0.0635        0.1400     +  0.0010  2.4512
      5        0.0552        0.1366     +  0.0010  2.4371
      6        0.0545        0.1337     +  0.0010  2.3827
      7        0.0402        0.1338        0.0010  2.4001
      8        0.0414        0.1379        0.0010  2.3948
      9        0.0430        0.1436        0.0010  2.4074
     10        0.0359        0.1494        0.0010  2.4269
Stopping since valid_loss has not improved in the last 5 epochs.
Migliori iperparametri: {'batch_size': 64, 'optimizer__lr': 0.001}
Miglior MSE: 0.10570659736792247
freeze_bert=False
Fitting 3 folds for each of 2 candidates, totalling 6 fits

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1619        0.1635     +  0.0000  4.6446
      2        0.1363        0.1549     +  0.0000  4.6990
      3        0.1131        0.1446     +  0.0000  4.7685
      4        0.0929        0.1402     +  0.0000  4.8261
      5        0.0928        0.1400     +  0.0000  4.7781
      6        0.0850        0.1400     +  0.0000  4.6289
```

```
      7        0.0764        0.1414        0.0000  4.7829
      8        0.0612        0.1430        0.0000  4.7679
      9        0.0633        0.1415        0.0000  4.8496
     10        0.0557        0.1423        0.0000  4.9470
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 1/3] END batch_size=16, optimizer__lr=1e-05;, score=-0.106 total time=
3.3min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight  | UNEXPECTED |  |
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
cls.seq_relationship.weight                 | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1501        0.1656     +  0.0000  4.5298
      2        0.1305        0.1596     +  0.0000  4.5633
      3        0.1081        0.1526     +  0.0000  4.6147
      4        0.0920        0.1497     +  0.0000  4.7114
      5        0.0878        0.1488     +  0.0000  4.7948
      6        0.0788        0.1500        0.0000  4.8067
      7        0.0633        0.1498        0.0000  4.8007
      8        0.0604        0.1488     +  0.0000  4.7832
      9        0.0558        0.1481     +  0.0000  4.5555
     10        0.0459        0.1472     +  0.0000  4.6379
     11        0.0506        0.1499        0.0000  4.7456
     12        0.0440        0.1478        0.0000  4.7625
     13        0.0371        0.1507        0.0000  4.7842
     14        0.0394        0.1500        0.0000  4.8768
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 2/3] END batch_size=16, optimizer__lr=1e-05;, score=-0.092 total time=
2.8min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
```

```
--------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
cls.seq_relationship.weight                 | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1605        0.1552     +  0.0000  4.5527
      2        0.1122        0.1437     +  0.0000  4.5811
      3        0.0912        0.1334     +  0.0000  4.6789
      4        0.0851        0.1280     +  0.0000  4.7336
      5        0.0687        0.1253     +  0.0000  4.7908
      6        0.0619        0.1224     +  0.0000  4.7662
      7        0.0580        0.1241        0.0000  4.7185
      8        0.0510        0.1239        0.0000  4.6717
      9        0.0520        0.1212     +  0.0000  4.7069
     10        0.0459        0.1228        0.0000  4.5577
     11        0.0423        0.1226        0.0000  4.5859
     12        0.0335        0.1210     +  0.0000  4.6654
     13        0.0415        0.1229        0.0000  4.7576
     14        0.0367        0.1255        0.0000  4.7797
     15        0.0387        0.1282        0.0000  4.8784
     16        0.0243        0.1274        0.0000  4.9145
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 3/3] END batch_size=16, optimizer__lr=1e-05;, score=-0.105 total time=
3.1min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                         | Status     |  |
--------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias    | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight      | UNEXPECTED |  |
cls.predictions.bias                        | UNEXPECTED |  |
cls.seq_relationship.bias                   | UNEXPECTED |  |
cls.predictions.transform.dense.bias        | UNEXPECTED |  |
cls.seq_relationship.weight                 | UNEXPECTED |  |
```

```
Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1731        0.1583     +  0.0000  4.6311
      2        0.1332        0.1482     +  0.0000  4.4859
      3        0.1075        0.1352     +  0.0000  4.5708
      4        0.0852        0.1335     +  0.0000  4.7271
      5        0.0895        0.1326     +  0.0000  4.7776
      6        0.0601        0.1329        0.0000  4.7288
      7        0.0578        0.1319     +  0.0000  4.7386
      8        0.0374        0.1302     +  0.0000  4.8301
      9        0.0408        0.1255     +  0.0000  4.6404
     10        0.0326        0.1321        0.0000  4.7044
     11        0.0313        0.1347        0.0000  4.7587
     12        0.0387        0.1480        0.0000  4.8340
     13        0.0274        0.1467        0.0000  4.9186
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 1/3] END batch_size=16, optimizer__lr=2e-05;, score=-0.089 total time=
5.1min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1615        0.1636     +  0.0000  4.5548
      2        0.1282        0.1519     +  0.0000  4.6623
      3        0.1087        0.1407     +  0.0000  4.7217
      4        0.0895        0.1288     +  0.0000  4.7618
      5        0.0778        0.1258     +  0.0000  4.8238
      6        0.0637        0.1302        0.0000  4.6714
```

```
      7        0.0553        0.1318        0.0000  4.6397
      8        0.0459        0.1333        0.0000  4.7339
      9        0.0457        0.1346        0.0000  4.8041
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 2/3] END batch_size=16, optimizer__lr=2e-05;, score=-0.099 total time=
2.5min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1774        0.1615     +  0.0000  4.5795
      2        0.1377        0.1578     +  0.0000  4.6557
      3        0.0999        0.1562     +  0.0000  4.7537
      4        0.0920        0.1569        0.0000  4.7647
      5        0.0742        0.1576        0.0000  4.8030
      6        0.0676        0.1643        0.0000  4.9521
      7        0.0523        0.1683        0.0000  4.9440
Stopping since valid_loss has not improved in the last 5 epochs.
[CV 3/3] END batch_size=16, optimizer__lr=2e-05;, score=-0.136 total time=
2.4min

Loading weights:   0%|          | 0/199 [00:00<?, ?it/s]

BertModel LOAD REPORT from: dbmdz/bert-base-italian-xxl-cased
Key                                        | Status     |  |
-------------------------------------------+------------+--+-
cls.predictions.transform.LayerNorm.bias   | UNEXPECTED |  |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |  |
cls.predictions.transform.dense.weight     | UNEXPECTED |  |
cls.predictions.bias                       | UNEXPECTED |  |
cls.seq_relationship.bias                  | UNEXPECTED |  |
cls.predictions.transform.dense.bias       | UNEXPECTED |  |
cls.seq_relationship.weight                | UNEXPECTED |  |
```

```
Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture;
not ok if you expect identical arch.

Pesi caricati.
  epoch    train_loss    valid_loss    cp      lr     dur
-------  ------------  ------------  ----  ------  ------
      1        0.1557        0.1703     +  0.0000  7.1036
      2        0.1363        0.1646     +  0.0000  7.2740
      3        0.1074        0.1619     +  0.0000  7.4694
      4        0.0890        0.1584     +  0.0000  7.3989
      5        0.0865        0.1582     +  0.0000  7.5211
      6        0.0748        0.1579     +  0.0000  7.6553
      7        0.0624        0.1565     +  0.0000  7.2207
      8        0.0566        0.1532     +  0.0000  7.4264
      9        0.0492        0.1530     +  0.0000  7.4430
     10        0.0471        0.1526     +  0.0000  7.4729
     11        0.0431        0.1575        0.0000  7.3423
     12        0.0390        0.1592        0.0000  7.4690
     13        0.0324        0.1578        0.0000  7.6808
     14        0.0316        0.1571        0.0000  7.5990
Stopping since valid_loss has not improved in the last 5 epochs.
Migliori iperparametri: {'batch_size': 16, 'optimizer__lr': 1e-05}
Miglior MSE: 0.10133973509073257
```

5 Valutazione dei risultati

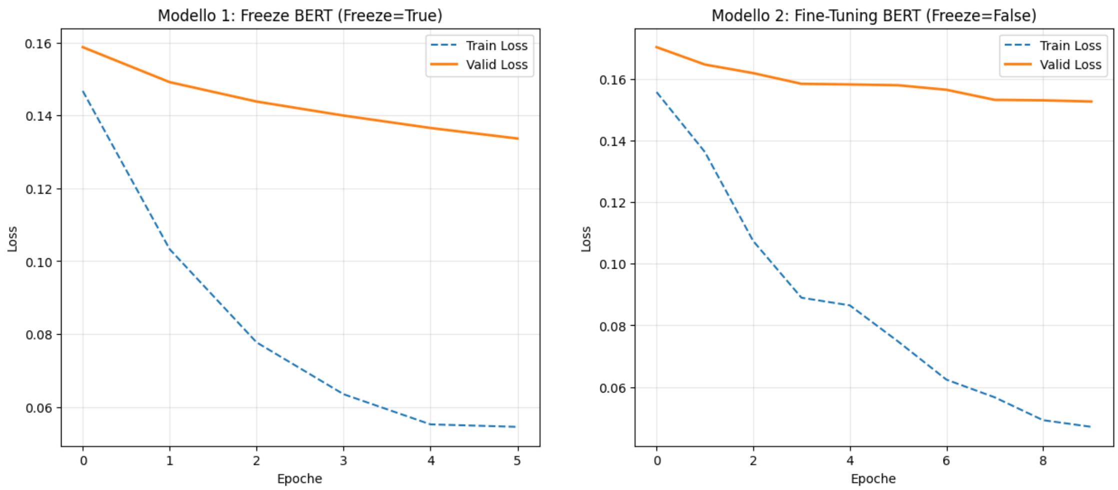
A questo punto vengono confrontati i valori del dizionario `risultati` e vengono stampati i grafici della loss e del MSE.

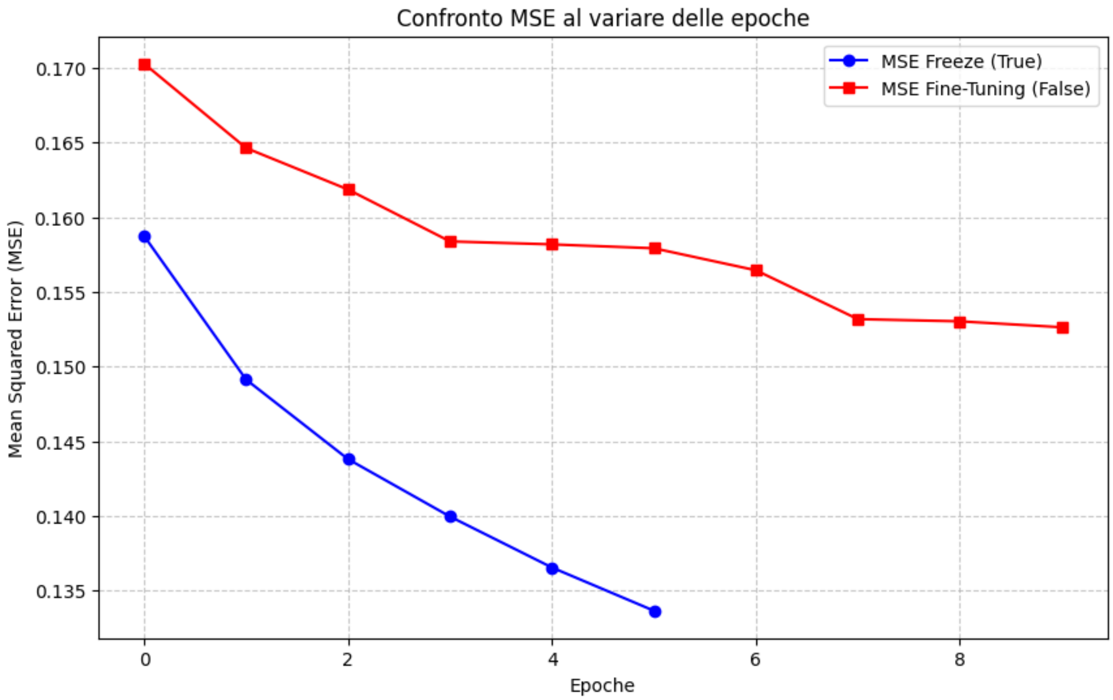
[15]:
```python
hist_freeze = risultati[True]['history']
hist_finetune = risultati[False]['history']

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(15, 6))
ax1.plot(hist_freeze[:, 'train_loss'], label='Train Loss', linestyle='--')
ax1.plot(hist_freeze[:, 'valid_loss'], label='Valid Loss', linewidth=2)
ax1.set_title("Modello 1: Freeze BERT (Freeze=True)")
ax1.set_xlabel("Epoche")
ax1.set_ylabel("Loss")
ax1.legend()
ax1.grid(True, alpha=0.3)
ax2.plot(hist_finetune[:, 'train_loss'], label='Train Loss', linestyle='--')
ax2.plot(hist_finetune[:, 'valid_loss'], label='Valid Loss', linewidth=2)
ax2.set_title("Modello 2: Fine-Tuning BERT (Freeze=False)")
ax2.set_xlabel("Epoche")
ax2.set_ylabel("Loss")
ax2.legend()
```

```
ax2.grid(True, alpha=0.3)
plt.show()

plt.figure(figsize=(10, 6))
plt.plot(hist_freeze[:, 'valid_loss'], label='MSE Freeze (True)', marker='o',␣
  ↪color='blue')
plt.plot(hist_finetune[:, 'valid_loss'], label='MSE Fine-Tuning (False)',␣
  ↪marker='s', color='red')
plt.title("Confronto MSE al variare delle epoche")
plt.xlabel("Epoche")
plt.ylabel("Mean Squared Error (MSE)")
plt.legend()
plt.grid(True, linestyle='--', alpha=0.7)
plt.show()
```





5.1 Selezione del modello

Il modello finale viene scelto in base al miglior valore di MSE ottenuto in validazione, privilegiando la capacità di generalizzazione rispetto alla semplice ottimizzazione della loss di training.

5.2 Inferenza su test set

Una volta selezionato il modello migliore, viene utilizzato per generare predizioni sul test set.

I risultati CSV e i pesi del modello vengono salvati per garantire riproducibilità e utilizzo futuro.

[16]:
```
mse_freeze = min(risultati[True]['valid_loss'])
mse_finetune = min(risultati[False]['valid_loss'])
print(f"Miglior MSE (Freeze=True):  {mse_freeze:.5f}")
print(f"Miglior MSE (Freeze=False): {mse_finetune:.5f}")

if mse_finetune < mse_freeze:
    print("\n>>> VINCITORE: Fine-Tuning (Freeze=False)")
    best_overall_model = risultati[False]['model']
    best_mode_str = "finetuning"
else:
    print("\n>>> VINCITORE: Freeze (Freeze=True)")
    best_overall_model = risultati[True]['model']
    best_mode_str = "freeze"
```

```python
TEST_PATH = "/content/drive/MyDrive/GSIdetect/dati/test.pkl" #"/content/drive/
↪MyDrive/Colab Notebooks/test.pkl"  # _pulito.pkl"

with open(TEST_PATH, "rb") as f:
    test_data = pickle.load(f)

X_test_text = test_data["text"]
if hasattr(X_test_text, 'reset_index'):
    X_test_text = X_test_text.reset_index(drop=True)

X_test = prepare_data(X_test_text, tokenizer, MAX_LEN)
print(f"len(X_test)={len(X_test)}")

test_preds = best_overall_model.predict(X_test)

df_results = pd.DataFrame({
    "text": X_test_text,
    "predicted_value": np.clip(test_preds.flatten(), 0.0, 1.0)
})

SAVE_CSV_PATH = f"/content/drive/MyDrive/GSIdetect/
↪risultati_test_BEST_{best_mode_str}.csv" #f"/content/drive/MyDrive/Colab␣
↪Notebooks/risultati_test_BEST_{best_mode_str}.csv"
SAVE_MODEL_PATH = f"/content/drive/MyDrive/GSIdetect/best_model_{best_mode_str}.
↪pkl" #f"/content/drive/MyDrive/Colab Notebooks/best_model_{best_mode_str}.
↪pkl"

df_results.to_csv(SAVE_CSV_PATH, index=False)
best_overall_model.save_params(f_params=SAVE_MODEL_PATH)

print(f"CSV salvato in: {SAVE_CSV_PATH}")
print(f"Modello salvato in: {SAVE_MODEL_PATH}")

print(df_results.head())
```

```
Miglior MSE (Freeze=True):  0.13366
Miglior MSE (Freeze=False): 0.15264

>>> VINCITORE: Freeze (Freeze=True)
len(X_test)=810
CSV salvato in: /content/drive/MyDrive/GSIdetect/risultati_test_BEST_freeze.csv
Modello salvato in: /content/drive/MyDrive/GSIdetect/best_model_freeze.pkl
                                                text  predicted_value
0  [Contesto: parlante donna] Vorrei dedicare un …         0.662107
1  Le gente appena ti vede ti chiede come mai non…         0.577766
2  Tenete duro ancora qualche giorno e i vostri f…         0.708190
3  Ma perché l'unico uomo in questa storia è piu …         0.536931
```

4	Sentivo qualcosa di speciale e sai, una donna …	0.569732